MAT 3103: Computational Statistics and Probability **Semester: Fall 2019-20**

Chapter 6: Stochastic Process

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Stochastic Process

It is a collection of random variable. A random variable X(t) is said to stochastic process if X takes different values at different points of time.

Examples:

(1) No. of signals sent from a station by time t.

(2) No. of e-mails received by a cyber cafe by time t.

6.0.1 Under what conditions a stochastic process becomes counting process?

A stochastic process $\{N(t), t \geq 0\}$ is said to be a counting process if N(t) represents the total number of events that have occurred up to time t. The counting process must satisfy the following:

(1) $N(t) \geq 0$.

(2) $N(t)$ is an integer.

(3) If $s \leq t$ then $N(s) \leq N(t)$.

(4) If $s < t$, then $N(t) - N(s)$ is the number of events occurred during $[s, t]$.

Example:

(1) Total no. of signals sent from a station by time t.

(2) Total no. of e-mails received by a cyber cafe by time t.

6.0.2 Under what conditions a stochastic process becomes Markov process?

A stochastic process in state i moves to state j with associated probabilities p_{ij} such that (i) $p_{ij} \geq 0$, (ii) $\sum_{j=0}^{\infty} p_{ij} = 1$, such a stochastic process is called Markov process.

6.0.3 Under what conditions a counting process becomes Poisson process?

A counting process N(t) is said to be a Poisson process if it occurs at Poisson rate λ and if it follows the followings:

(1) $N(0) = 0$

(2) N (t) increases independently

(3) The probability distribution of $N(t)$ is $P[N(t)] = \frac{e^{-\lambda t} \times (\lambda t)^n}{n!}$

6.0.4 Distribution of inter arrival time in case of Poisson Process:

Let T_1 be the time for the occurrence of first event. Now, if $n > 1, T_n$ is the elapsed time between the occurrence of $(n-1)th$ and nth event occurring after some time t.

$P[N(0) = 0] = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = \ e^{-\lambda t}$

$P(T > t) = \ e^{-\lambda t}$

The probability function of T is as $f(T) = \lambda \ e^{-\lambda t}$

Mean, $E(T) = 1/\lambda$

The total time until the occurrence of the nth event is $S_n = T_1 + T_2 + .. + T_n$

We need expected total time until the occurrence of nth event

$E(S_n) = E(T_1 + T_2 + .. + T_n) = E(T_1) + E(T_2) + .. + E(T_n) = n/\lambda$

Example 1: Customers enter to a mobile repairing shop from 9 am to 10 am and 10 am to 11 am with probability 0.6. They also enter to the shop from 10 am to 11 am but not from 9 am to 10 with probability 0.5. Find the probability that the customers will enter to the shop up to 1 pm starting from 9 am.

Solution: Our transition probability matrix:

$$P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix}$$

The process of entrance will move to 4 steps from 9 am to 1 pm. We need P^4_{00} in P^4 .

$$P^2 = \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix} \begin{bmatrix} 0.56 & 0.44 \\ 0.55 & 0.45 \end{bmatrix} = \begin{bmatrix} 0.5556 & 0.4444 \\ 0.5555 & 0.4445 \end{bmatrix}$$

The required probability is 0.5556.

Example 2: Emails enter to a cyber caf after several stages. An email sent from a cyber caf reaches to the next stage with probability 0.7. Find the probability that an email sent from cyber cafe reaches to another cyber caf to the address at 4th stage.

Our transition probability matrix:

$$P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

To reach its destination at 5th step, it needs further (4-1) = 3 steps. We need P_{00}^3 in P^3 .

$$P^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.532 & 0.468 \\ 0.468 & 0.532 \end{bmatrix}$$

The required probability is 0.532.

Example 3: An electronic device sends signals. The signals reach to the destination at a Poisson rate λ=5 per minute. Find the probability that the elapsed time between the entrance of 5th and 6th signal is (i) more than 1 minute, (ii) less than half a minute, (iii) 0.5 to 1 minute. Find the expected time until the 10th signal reaches to the destination.

Solution: (a) Let T be the elapsed time between the entrance of (n-1)th and nth signal and S_n be the waiting time until the nth signal reaches to the destination. We need

$P(T > 1) = \ e^{-\lambda t} = \ e^{-5\times 1} = 0.00674$

$P(T < 0.5) = 1 - \ e^{-\lambda t} = 1 - \ e^{-5\times 0.5} = 0.91792$

$P(0.5 < T < 1) = \ e^{-\lambda t_1} - \ e^{-\lambda t_2} = \ e^{-5\times 0.5} - \ e^{-5\times 1} = 0.08208 - 0.00674 = 0.07534$

$E(S_n) = n/\lambda = 10/5 = 2$

6.1 Exercise

1. Write down some examples of stochastic process and counting process.

2. Under what conditions a stochastic process becomes counting process; a counting process becomes Poisson process? What are the mean and variance of the distribution of waiting time?

3. Customers enter to a mobile repairing shop from 9 - 10 and from 10 - 11 with probability 0.6. Customers also enter to the shop from 10 - 11 but not from 9 - 10 with probability 0.8. Find the probability that the customers will enter to the cyber caf up to 2 pm.

4. Emails enter to a cyber caf after several stages. An email sent from a station reaches to the next stage with probability 0.8. Find the probability that an email sent from cyber cafe reaches to another cyber caf to the address at 5th stage.

5. E. mails enter to a cyber caf at a Poisson rate $\lambda = 2$ per minute. Find the probability that the elapsed time between the entrance of 10th and 11th mail is (i) more than 1 minute, (ii) less than 2 minutes, (iii) between 1 to 2 minutes. Find the expected time until the entrance of 10th mail.

6. The customer enters to a bank at Poisson rate $\lambda = 25$ per hour. Find the probability that the elapsed time between the entrance of 10th and 11th customer is (i) less than 2minutes, (ii) more than 5 minutes, (iii) 3 to 5 minutes. Find the expected time until the entrance of 50th customer.